

Roman Urdu first-stage retrieval for Urdu news: dense and hybrid candidates after a frozen lexical baseline

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Abstract

Ordinary Roman Urdu remains the main failure mode of a previously frozen script-aware BM25 retriever (M0) on Urdu news. M0 recovers title-derived known items when Roman queries match its own romanization family, but sealed ordinary-Roman tests leave a first-stage matching gap. This paper reports ULTRA v2 (Program B): a preregistered evaluation of first-stage methods on a new Roman known-item (KN) TRAIN+DEV population, without rewriting M0’s official scores and without opening the sealed TEST partition.

On the frozen expanded set ($n = 80$; 51 human-written and 29 LLM-drafted, human-reviewed queries), dense retrieval with multilingual-e5-small achieved the best ExactSource Hit@5 ($20/80 = 25\%$; 95% Wilson CI 0.168–0.355). Hybrid reciprocal-rank fusion of Method-D BM25 and dense ($k = 60$) achieved the best Hit@50 ($40/80 = 50\%$; 95% CI 0.393–0.607). Against a preregistered development stretch target of approximately 80% ExactSource Hit@5, the best observed Hit@5 was 25% (55 percentage points short). Method-D alone remained near floor ($5/80$ Hit@5).

A diagnostic of the 31 candidate-generation failures—queries whose gold never entered the BM25UNG3UDense Top-50—found Covered 17 / Partial 14 / Uncovered 0, with Potential_New_Mechanism = Yes on 0/31. Residual misses were heterogeneous and mapped onto existing method families. No common previously unmodeled mechanism was identified among the 31 analyzed candidate-generation failures. The contribution is empirical and diagnostic, not a new retrieval algorithm. Program B results must not be averaged with M0’s 87.18% / 67.50% / 57.50% headlines.

Introduction

Urdu news search is bilingual in practice. Many users type Perso-Arabic script; many others type informal Roman Urdu [1–3]. Shared evaluation for Urdu NLP and IR is still thinner than for English [4]. Ranking collections such as CURE [5] and Urdu MS MARCO [6] exist, but they do not freeze a script-conditional lexical baseline and then measure ordinary Roman known-item recovery on a separate TRAIN/DEV/TEST program. Most Roman Urdu work remains classification—sentiment, offensive language—rather than news search [1, 2]. Multilingual dense retrievers can lose effectiveness on languages outside the pretraining head [7], even when scaled multilingual encoders help related tasks [8]. Users of lower-resource varieties are also often pushed toward high-resource query forms [9].

The ULTRA dual-embedding Urdu news architecture [10] originally switches on a static length threshold. Length does not decide which script index to open. English work on choosing sparse versus dense retrieval [11] or adapting retrieval depth [12] is related engineering context, not the protocol evaluated here. Okapi BM25 remains a strong lexical baseline [13]. Reciprocal rank fusion (RRF) is a standard way to combine ranked lists without score calibration [14].

A frozen lexical system called M0 routes Urdu and Roman queries to separate BM25 indexes and reports three official metrics that answer different questions: ExactSource Hit@5 of 68/78 (87.18%) on a development/validation known-item pool, 27/40 (67.50%) on a sealed known-item set K, and human Success@5 of 23/40 (57.50%) on a sealed naturalistic set U. Native-script Urdu search on that collection is often useful. Ordinary Roman Urdu is not. On sealed K, ordinary Roman titles recovered the source in 1 of 12 cases; on sealed U, Roman queries succeeded in 6 of 18. Those sealed sets were burned for further M0 tuning once retrieved and labeled. They cannot serve as an honest test bed for a later Roman-repair program.

The high development score mixes a different Roman construction. Development Roman strings were pipeline `title_roman` forms—dictionary reverse lookup plus a closed character table—not chat-style spelling. Method D, the romanized-document BM25 path that M0 uses for ROMAN queries, was selected on that family [13]. Treating 87.18% as evidence that Roman Urdu search is solved would misread the freeze. The scientific gap left for follow-on work is first-stage recovery of ordinary Roman known items on a fresh population, under preregistration, with TEST sealed until a single candidate is frozen.

ULTRA v2 (Program B) is that follow-on. It keeps the same 111,860-article corpus and the same frozen M0 code as a baseline, builds a new TRAIN/DEV/TEST benchmark, and scores Roman KN ExactSource Hit@ k only on TRAIN+DEV. The v2 protocol also recorded a preregistered development stretch target of approximately 80% ExactSource Hit@5 on the Roman KN TRAIN+DEV population; that figure is an internal planning bar, not an external leaderboard or community standard. This manuscript reports the frozen Program B outcome, including distance from that stretch target as secondary context, and a post-freeze diagnostic of residual candidate-generation failures. We do not claim state of the art against CURE or Urdu MS MARCO [5, 6]. This paper does not propose a new retrieval algorithm. It provides a preregistered empirical and diagnostic evaluation of first-stage recovery for ordinary Roman Urdu known-item queries against a frozen lexical baseline, quantifies the gap relative to an aligned development setting, and shows that residual candidate-generation failures are heterogeneous and map onto existing method families rather than a single newly identified mechanism.

We ask four empirical questions on Roman KN TRAIN+DEV only.

Research questions

- **RQ1.** How often does frozen Method-D BM25 recover the ExactSource document for new human-written Roman KN queries ($n = 51$), relative to the development `title_roman` setting where Method D was selected?
- **RQ2.** Do controlled first-stage alternatives—fallback romanization, character n -gram overlap (NG3), dense e5-small, two-way hybrid RRF, letter-name expansion, three-way RRF, Wikipedia-title expansion, and cascade variants—improve ExactSource Hit@5 or Hit@50 on that population under preregistered stop rules?

- **RQ3.** Do the selected methods’ rates hold under a frozen population expansion to $n = 80$ when scored without retuning?
- **RQ4.** How large is the remaining first-stage retrieval gap after the preregistered methods are frozen?

Materials and methods

Research design

Program B is an offline known-item evaluation of first-stage retrieval methods on Roman Urdu queries over the same hashed Urdu news collection used by M0. Each scored phase was preregistered before ranking. Negative outcomes closed methods under written stop rules; they did not trigger silent retuning. The sealed TEST partition of the v2 benchmark was never opened for retrieval in any Program B phase. Program A headlines (68/78, 27/40, 23/40) are historical context only and are not recomputed or averaged here.

The primary metric is ExactSource Hit@ k : 1 if and only if the designated source document identifier appears in the retrieved top k . Official cutoffs reported throughout are Hit@1, Hit@5, Hit@10, and Hit@50, plus mean reciprocal rank (MRR). Human Success@5 is not scored in this paper: the naturalistic (NL) track was not labeled for Program B.

Corpus and frozen lexical baseline

The collection is `data/clean_articles.csv`: 111,860 articles; SHA-256 8992a6acca3459eea17a7d7356dd490445daa00b958eab765713853c97a9f231. The local precursor is schema- and size-consistent with the third-party Urdu News Dataset 1M compilation [15,16]. The Roman dictionary used by Method D remains `models/roman_urdu_dict_expanded.json` (198 keys; SHA-256 30c3f61a64ec641abbb3acdbc7a8bcaf197f0238f1bf9e76c2c7ce8e590f86a3). Method-D BM25 (R2-B0) is imported from the frozen M0 implementation and is not edited in place. BM25 uses $k_1 = 1.5$ and $b = 0.75$ [13]. Lists are retrieved to depth 50.

Benchmark population

Roman KN queries are known-item strings with a single `source_doc_id`. The original Program B TRAIN+DEV Roman KN set has $n = 51$ human-written queries (`writer_id = W1`). Phase 13 froze an expanded TRAIN+DEV set of $n = 80$: the original 51 plus 29 new queries (KN091–KN119) that were LLM-drafted and human naturalness-reviewed (`writer_id = LLM1`). Authorship of those 29 must not be blurred with the human-written 51. The TEST directory exists and remains sealed; Program B runners hard-refuse TEST paths. NL queries were collected under the same benchmark design but are not scored here.

Methods compared (Phases 2–10b)

Methods were introduced in sequence. Each Hit@ k claim below cites the frozen Phase 14 consolidation (S1 File), which supersedes scattered per-phase write-ups for Program B citation.

R2-B0 (Method-D BM25). Baseline Roman path: typed Roman query against documents romanized with the reverse 198-key dictionary and the Phase 2 character table.

R2-1. Controlled fallback romanizer on the document side. Preregistered as a close of that direction if Hit@50 did not improve.

R2-NG3 / Phase 5 NG3. Character n -gram overlap matching as a complementary lexical channel.

Dense. `intfloat/multilingual-e5-small` first-stage dense retrieval over the news collection (Program B build; not the abandoned M0 CPU-time gate for a full e5 index in Program A). Cross-lingual dense retrieval is known to be fragile for some low-resource settings [7,8].

Hybrid. Reciprocal rank fusion of Method-D BM25 and Dense with RRF constant $k = 60$ [14].

Phase 7 letter-name. Acronym-to-Urdu letter-name expansion as a query treatment.

Phase 8 three-way RRF. Unweighted RRF of BM25, Dense, and NG3 ($k = 60$) [14]. 93

Phase 9. Wikipedia-title entity resource: expand queries using exact title keys under a frozen matcher, 94
then retrieve (S4 File). 95

Phase 10 Option B. Hybrid-first cascade that fills from an auxiliary list (S5 File). 96

Phase 10b. Tail-preserving cascade variant; stopping rule: stop further Option-B variants if the result 97
is identical to Hybrid with no net new recoveries (S5 File). 98

Phases 11–14 (gating, population, consolidation) 99

Phase 11. Inspection of remaining vocabulary / paraphrase dual-misses where Dense also fails badly 100
(KN020, KN028, KN036, KN041, KN042). An Option-1 probe tested whether frozen Phase 9 resources 101
could inject candidate lists for acronym and institutional misses (including CPEC and SBP title lookups). 102
No viable candidate-generation path from existing frozen resources was found; Option 2 (building a new 103
bilingual glossary) was not authorized after that no-go. Phase 11 is therefore not a Hit@ k row (S1 File). 104

Phase 12 (reranking). Skipped under a gating rule: reranking presupposes a pool that already 105
contains the gold more often. Phase 9’s gain remained small at both $n = 51$ and $n = 80$; Phase 10 106
regressed Hit@50 relative to Hybrid; Phase 10b equalled Hybrid. The program did not spend a phase 107
reordering a pool that was not genuinely expanded (S1 File). 108

Phase 13. Population freeze to $n = 80$ and confirmatory scoring of frozen methods without retuning. 109
Scoring was locked by preregistration (SHA-256 prefix 55788660...) before the scoring runner executed. The 110
 $n = 51$ subset of the $n = 80$ run was required to exact-match frozen Phase 2–10b cells (gate PASS) (S2 111
File). 112

Phase 14. Consolidation report: single source of truth for Program B claims (file SHA-256 113
0cf7c9a928e9d9e734293d18c1d8fbab8c167775340a880d76c0dcaeea862f7b) (S1 File). 114

Candidate-generation failure diagnostic 115

After Phase 14 froze the Hit@ k tables, we ran a separate diagnostic on the 31 TRAIN+DEV queries whose 116
ExactSource gold never appeared in the union of Method-D BM25, NG3, and Dense Top-50 lists 117
(candidate-generation failures). Each case was compared against the gold document text and tagged for 118
coverage by existing method families: transliteration, entity linking, acronym expansion, character n -gram 119
overlap, dense multilingual matching, and query expansion. Tags were assigned as Covered, Partial, or 120
Uncovered for existing-method coverage, plus a binary Potential_New_Mechanism flag. This pass did not 121
retune retrieval methods, did not open TEST, and did not alter any frozen Program B count. 122

Confidence intervals 123

For the main $n = 80$ headline rates we report binomial 95% Wilson score confidence intervals. Intervals are 124
given for Dense Hit@5 (20/80), Hybrid Hit@5 (17/80), and Hybrid Hit@50 (40/80). Dense and Hybrid are 125
scored on the same queries, so comparisons between them are paired; we report the intervals as descriptive 126
uncertainty around each rate and do not claim a formal superiority test. 127

Integrity constraints 128

TEST access is recorded as false in controlled-experiment summaries for Phases 2–13 and in the post-freeze 129
diagnostic. Unsupported methods were closed rather than retuned after negative results (R2-1, Phase 7, 130
Phase 8; Phase 10 Option-B stopping after 10b; Phase 11 Option 2 skipped; Phase 12 skipped). Program B 131
numbers must not replace or be averaged into Program A headlines. 132

Ethics 133

No search-user logs or identifiable participant records were collected for Program B scoring. KN labels are 134
ExactSource identifiers assigned at query writing, not graded usefulness judgments. This study is an offline 135
retrieval evaluation using an existing, previously compiled news corpus (`data/clean_articles.csv`) and 136

did not involve human participants, identifiable individuals, or personal data collection; institutional ethics review was therefore not required.

Results

Table 1 states the Program B headline, including distance from the preregistered development stretch target. Tables 2 and 3 give the consolidated ExactSource counts for $n = 51$ and $n = 80$. Table 4 separates the human-written (W1) and LLM-drafted (LLM1) slices for the main Dense and Hybrid rates. All counts are copied from the Phase 14 consolidation of frozen JSON summaries (S1 File); they were not recomputed for this manuscript.

Table 1. Program B headline versus preregistered development stretch target. ExactSource on Roman KN TRAIN+DEV. Do not average with M0’s 87.18% / 67.50% / 57.50%.

Quantity	Result
Best Hit@5 ($n = 80$)	Dense: 20/80 = 0.250 (95% CI 0.168–0.355)
Best Hit@50 ($n = 80$)	Hybrid/10b (tied): 40/80 = 0.500 (95% CI 0.393–0.607)
Hybrid Hit@5 ($n = 80$)	17/80 = 0.2125 (95% CI 0.137–0.314)
Phase 10 Option B Hit@50	39/80 (worse than Hybrid/10b)
Method-D Hit@5 ($n = 51 / 80$)	4/51 = 0.0784; 5/80 = 0.0625
Preregistered stretch (~80% Hit@5)	Best Hit@5 25%; 55 points short
Hit@50 vs 80% bar at depth 50	Peak 50%; still 30 points short
Program B vs stretch	Stretch target not reached

Counts from S1 File. Dense Hit@5 MRR = 0.1718. Confidence intervals are 95% Wilson score intervals. The ~80% figure is a preregistered development stretch target, not an external benchmark.

Human-written Roman KN ($n = 51$)

Method-D BM25 recovered the source in the Top-5 for 4 of 51 queries (Table 2). Dense e5-small recovered 15/51 at Hit@5 and 22/51 at Hit@50 (MRR 0.1726). Hybrid RRF scored 11/51 Hit@5 and 25/51 Hit@50 (MRR 0.1422). NG3 scored 6/51 Hit@5 and 8/51 Hit@50. Phase 9 Wikipedia titles scored 5/51 Hit@5 and 10/51 Hit@50. Phase 10 Option B matched Hybrid through Hit@10 but fell to 24/51 Hit@50. Phase 10b matched Hybrid exactly at Hit@50 (25/51).

R2-1 failed its improvement gate: Hit@50 stayed 6/51 while Hit@5 fell from 4 to 3; the direction was closed as not supported. Phase 7 letter-name scored 0/51 Hit@5 and 1/51 Hit@50 (MRR 0.0007) and was unsupported. Phase 8 three-way RRF scored 8/51 Hit@5 and 24/51 Hit@50—worse Hit@5 than two-way Hybrid (11)—and was unsupported.

Expanded population ($n = 80$) without retuning

Confirmatory scoring on the frozen $n = 80$ set left the ranking story intact (Table 3). Dense remained best at Hit@5 (20/80; 95% Wilson CI 0.168–0.355) and MRR (0.1718). Hybrid Hit@5 was 17/80 (95% CI 0.137–0.314). Hybrid and Phase 10b tied at Hit@50 (40/80; 95% CI 0.393–0.607). Phase 10 Option B remained one Hit@50 behind (39/80). Method D stayed near floor (5/80 Hit@5). Dense versus Hybrid comparisons are paired over the same queries; the intervals describe uncertainty around each rate and are not presented as a formal paired significance test. Methods never re-scored on $n = 80$ (R2-1, Phase 7, Phase 8) are marked n/a and retain their $n = 51$ unsupported decisions.

Table 4 makes the authorship split visible. On human-written queries (W1, $n = 51$), Dense Hit@5 was 15/51, Hybrid Hit@5 was 11/51, and Hybrid Hit@50 was 25/51. On the LLM-drafted, human-reviewed slice (LLM1, $n = 29$), the corresponding rates were 5/29, 6/29, and 15/29. The totals match the frozen

Table 2. ExactSource on Roman KN TRAIN+DEV, $n = 51$ (human-written, W1). Decisions are frozen Program B outcomes.

Method	Decision	H@1	H@5	H@10	H@50	MRR
Method-D BM25	Baseline	1	4	4	6	0.0375
R2-1 fallback	Not supported	1	3	4	6	0.0358
NG3	Partially supported	3	6	6	8	0.0750
Dense e5-small	Partially supported	5	15	16	22	0.1726
Hybrid RRF $k=60$	2-way baseline	3	11	19	25	0.1422
Phase 7 letter-name	Unsupported	0	0	0	1	0.0007
Phase 8 3-way RRF	Unsupported	4	8	14	24	0.1310
Phase 9 titles	Partially supported	1	5	6	10	0.0478
Phase 10 Option B	Partially supported	3	11	19	24	0.1418
Phase 10b	Partially supported	3	11	19	25	0.1422

Source: Phase 14 consolidation of frozen JSON summaries (S1 File).

Table 3. ExactSource on Roman KN TRAIN+DEV, $n = 80$ (51 W1 + 29 LLM1). No retuning. $n = 51$ subset gate PASS against frozen Phase 2–10b cells.

Method	H@1	H@5	H@10	H@50	MRR
Method-D BM25	1	5	5	13	0.0336
NG3	4	10	13	19	0.0766
Dense e5-small	9	20	23	36	0.1718
Hybrid RRF $k=60$	5	17	25	40	0.1376
Phase 9 titles	1	9	10	19	0.0520
Phase 10 Option B	5	17	25	39	0.1373
Phase 10b	5	17	25	40	0.1376
R2-1 / Ph. 7 / Ph. 8	n/a	n/a	n/a	n/a	n/a

n/a: not re-scored on $n = 80$. Counts from S1 File / S2 File.

$n = 80$ cells in Table 3. The LLM1 slice is reported for inspectability; it was not used for method selection or retuning (S2 File).

Table 4. Authorship-slice ExactSource rates for Dense and Hybrid. W1 = human-written; LLM1 = LLM-drafted and human-reviewed. Totals match Tables 2–3.

Slice	n	Dense Hit@5	Hybrid Hit@5	Hybrid Hit@50
W1 (human-written)	51	15/51	11/51	25/51
LLM1 (LLM-drafted, human-reviewed)	29	5/29	6/29	15/29
Total TRAIN+DEV	80	20/80	17/80	40/80

Counts from S1 File / S2 File. LLM1 rates are descriptive and were not a tuning target.

Skipped and closed phases

Phase 11 did not produce a Hit@ k improvement path from frozen resources for the five deep vocabulary dual-misses (Dense gold ranks 110, 523, 1674, 11599, and 469 for KN020, KN028, KN036, KN041, and KN042). Phase 12 reranking was not run. Those absences are results of gating, not omitted negative trials (S1 File).

Candidate-generation failure coverage

On the $n = 80$ population, 28/80 queries were recovered in at least one expert Top-5 (SUCCESS), 21/80 had gold in the BM25 $\cup$ NG3 $\cup$ Dense Top-50 but not Top-5 (RANKING_FAILURE), and 31/80 had gold absent from that Top-50 union (CG_FAILURE). Among those 31 CG failures, existing-method coverage was Covered 17, Partial 14, and Uncovered 0. Potential_New_Mechanism was Yes for 0/31. The misses fragmented across institutional acronym/alias forms, brand or title script bridges, person/geo name variants, English jargon composed with entity or event cues, underspecified or descriptive referents, and residual bilingual paraphrase. No common previously unmodeled mechanism was identified among the 31 analyzed candidate-generation failures.

Discussion

Program B answers RQ1 bluntly. Method D, which scored 22/23 on development `title_roman` under Program A method selection, scores 4/51 Hit@5 on new Roman KN. The lexical romanized-document path does not transfer to this ordinary Roman known-item population. That matches M0’s sealed K/U Roman failures, and it is cleaner evidence here because the queries were written for a new benchmark rather than after M0 was burned.

RQ2 splits into what moved and what did not. Dense e5-small is the strongest single first-stage method: largest Hit@5 and MRR at both $n = 51$ and $n = 80$. It recovers some paraphrase and entity cases that Method D cannot bridge, while still missing a hard vocabulary tail. Hybrid RRF is the strongest deployable depth-50 pool among the fusions tried. It restores lexical recalls that Dense drops, at some cost to Hit@5 relative to Dense alone. Phase 9 helps a little on entity/title cases but does not cover FBR/CPEC/ADB/JLO-class acronyms absent as exact Wikipedia title keys under the frozen rule. R2-1 did not build a BM25 bridge for the residual miss room. Letter-name expansion fired without recovering the hypothesized organizations in the Top-50. Unweighted three-way RRF diluted Hybrid ranks (Hit@5 11 $\rightarrow$ 8 at $n = 51$): union membership is not a ranked Top- k . Phase 10’s cascade evicted Hybrid-tail golds faster than it added recoveries; Phase 10b removed those regressions and collapsed to Hybrid, which triggered the written stop on further Option-B variants.

RQ3 holds for the ranking story under population growth. Expanding from 51 to 80 without retuning did not invent a new winner. Dense still leads Hit@5; Hybrid/10b still lead Hit@50; Method D remains near floor. Table 4 shows that the W1 and LLM1 slices are both hard; the LLM1 rates are descriptive only. The $n = 51$ subset gate PASS is the integrity check that the expansion did not silently change frozen cells.

RQ4 asks how large the remaining first-stage gap is after the preregistered methods freeze. On $n = 80$, best Hit@5 is 25% (Dense) and best Hit@50 is 50% (Hybrid/10b). Against a preregistered development stretch target of approximately 80% ExactSource Hit@5, the best observed Hit@5 was 25% (55 percentage points short). Even Hit@50 peaks at 50%. Program B freezes that shortfall as a result. The stretch target is secondary context for reading the size of the gap; it is not treated as an external benchmark.

The CG-failure diagnostic adds a final layer to that shortfall. The 31 Top-50 misses are real, but they are not one unexplained bridge. Under the Covered / Partial / Uncovered coding, 17 were Covered and 14 Partial; none were Uncovered, and none carried Potential_New_Mechanism = Yes. Residual difficulty looks like a mixture of known IR problems—transliteration and script variants, entity linking, acronym or institutional alias expansion, character n -gram matching, dense paraphrase, and query expansion. No common previously unmodeled mechanism was identified among the 31 analyzed candidate-generation failures. That finding does not make the Hit@5 shortfall smaller. It does change how the shortfall should be read: stacking familiar bridges may still help operationally, but the evidence from these TRAIN+DEV misses does not support claiming a new Roman Urdu retrieval mechanism.

This paper does not propose a new retrieval algorithm. It provides a preregistered empirical and diagnostic evaluation of first-stage recovery for ordinary Roman Urdu known-item queries against a frozen lexical baseline, quantifies the gap relative to an aligned development setting, and shows that residual candidate-generation failures are heterogeneous and map onto existing method families rather than a single newly identified mechanism. It is not a replacement for M0’s three official metrics. Relative to CURE and

Urdu MS MARCO [5, 6], the protocol here is news known-item ExactSource on ordinary Roman queries, not a passage-ranking leaderboard entry.

Limitations

All Hit@ k figures are TRAIN+DEV only. TEST is sealed and unused; no claim about held-out generalization is made. NL usefulness is unscored. The $n = 29$ expansion queries are LLM-drafted with human review; they are disclosed and sliced in Table 4, but they are not interchangeable with the human-written 51. Dense and Hybrid were not deployed as a replacement production system; Program A M0 remains the frozen lexical baseline for its own evaluations. The CG-failure tags are case-level judgments against gold documents on TRAIN+DEV; they are not a second retrieval experiment and should not be read as proof that any one existing method would have recovered all Covered cases into Top-5. Results are specific to the 111,860-article Urdu news collection used here and should not be assumed to generalize automatically to social media, web search, or other corpora. The evaluation is known-item (ExactSource) recovery, which is not identical to usefulness for real users. We do not assert an external acceptance letter or DOI for Program A.

Conclusion

On a new Roman known-item TRAIN+DEV population, frozen Method-D BM25 recovers few ExactSource documents (4/51 Hit@5; 5/80 after expansion). Dense multilingual-e5-small is the strongest single first-stage method (20/80 Hit@5; 95% CI 0.168–0.355). Hybrid RRF ($k = 60$) and an equivalent tail-preserving cascade are the strongest depth-50 pools tried (40/80 Hit@50; 95% CI 0.393–0.607). Letter-name expansion, unweighted three-way fusion, and cascade fill that evicts Hybrid-tail golds do not help. Wikipedia-title expansion helps narrowly. Reranking was not justified after candidate generation failed to expand the pool. Against a preregistered development stretch target of approximately 80% ExactSource Hit@5, the best observed Hit@5 was 25% (55 percentage points short). Among the 31 residual candidate-generation failures, coverage was Covered 17 / Partial 14 / Uncovered 0, with no Potential_New_Mechanism = Yes case. No common previously unmodeled mechanism was identified among the 31 analyzed candidate-generation failures. TEST was never opened. This paper does not propose a new retrieval algorithm; it offers a preregistered empirical and diagnostic map of first-stage Roman Urdu known-item recovery against a frozen lexical baseline.

Supporting information

S1 File. Phase 14 final report (Program B consolidation). Single source of truth for Program B Hit@ k claims, skipped-phase record (Phases 11–12), integrity table, and artifact index. Repository path: `experiments/ultra_v2/PHASE14_FINAL_REPORT.md` (SHA-256 `0cf7c9a928e9d9e734293d18c1d8fbab8c167775340a880d76c0dcaeea862f7b`). Companion hash file: `PHASE14_FINAL_REPORT.sha256`.

S2 File. Phase 13 population freeze and confirmatory scoring. Freeze manifest (51 W1 + 29 LLM1), scoring preregistration (SHA-256 prefix 55788660...), scoring summary JSON, and per-query ranks CSV. Paths under `experiments/ultra_v2/phase13_population/artifacts/`.

S3 File. Phases 2–8 preregistrations and controlled-experiment summaries. Method-D baseline, R2-1, Dense, Hybrid, NG3, letter-name, and three-way RRF preregistrations plus frozen JSON summaries listed in the Phase 14 artifact index.

S4 File. Phase 9 Wikipedia-title entity resource. Preregistration, controlled-experiment report, and frozen summary JSON under `experiments/ultra_v2/phase9_entity_resource/`.

S5 File. Phase 10 / 10b cascade bundle. Option B and tail-preserve preregistrations, controlled-experiment reports, and frozen summary JSON under experiments/ultra_v2/phase10_cascade/.

S6 File. Corpus and dictionary freeze hashes. Corpus SHA-256 8992a6acca3459eea17a7d7356dd490445daa00b958eab765713853c97a9f231; dictionary SHA-256 30c3f61a64ec641abbb3acdbc7a8bcaf197f0238f1bf9e76c2c7ce8e590f86a3 (models/roman_urdu_dict_expanded.json, 198 keys).

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